Batterjee Medical College
For Science and Technology

The Posterior Pituitary Disorders

Internal Medicine Department

Learning Objectives

Knowledge

- Relate pathophysiological basics and etiology of posterior pituitary gland disorders
- Describe epidemiology, manifestations, complications and management of posterior pituitary gland disorders

Skills

- Interpret clinical manifestations and investigations of posterior pituitary gland disorders to formulate reasonable diagnosis.
- Apply evidence-based management plan for posterior pituitary gland disorders
- Distinguish patients in emergency situations to formulate a comprehensive management plan; ensuring keeping patients in normal homeostasis

Main Subtopics

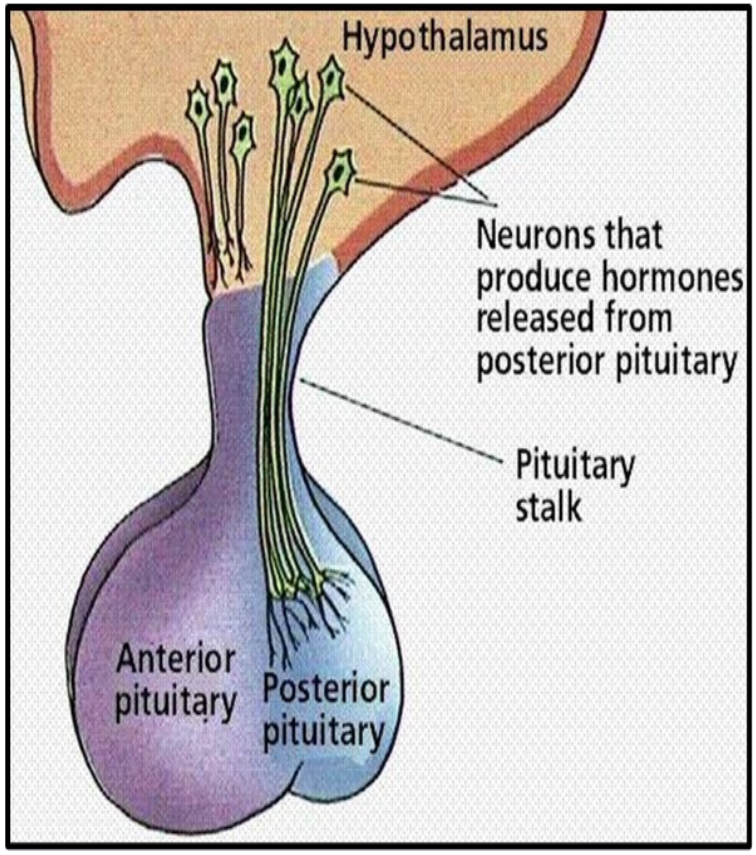
- Diabetes insipidus (clinical picture, investigations, differentiation of central versus nephrogenic DI)
- Treatment of diabetes insipidus (hormonal replacement, non-hormonal agents, treatment of nephrogenic DI)
- Differential diagnosis of polyuria (renal, endocrine, functional, and miscellaneous causes)
- SIADH (causes, diagnostic criteria)
- Diagnostic approach to hyponatraemia (osmolality-based algorithm, volume status assessment)

POSTERIOR PITUITARY

- Vasopressin (ADH) & oxytocin are secreted by the supra-optic & paraventricular nuclei of the hypothalamus respectively & migrate along the axons of these nuclei to the posterior pituitary.

➢ Oxytocin: stimulates the myo-epithelial cells of mammary gland to cause milk ejection (milk-letting effect).

➢ Vasopressin (Arginine vasopressin, (AVP) or Antidiuretic hormone (ADH) : It increases absorption of water in collecting tubules (also it constricts smooth muscles in the arterioles).



Control of secretion of ADH

1.Osmolar stimulation of centres within the hypothalamus when osmoregulation is required i.e. (↑ADH release when plasma osmolality ↑ & vice versa).

2.Reflex stimulation by ↓ plasma volume: ↓plasma volume →↓ inhibitory impulses from left atrium on ADH release from the hypothalamus.

3.Hypotension : ++ baroreeeptors (of carotid sinsus & aortic arch) → ↑↑ ADH release.

4.Higher centers stimulated by psychological factors & stress.

5.Pharmacological influence :

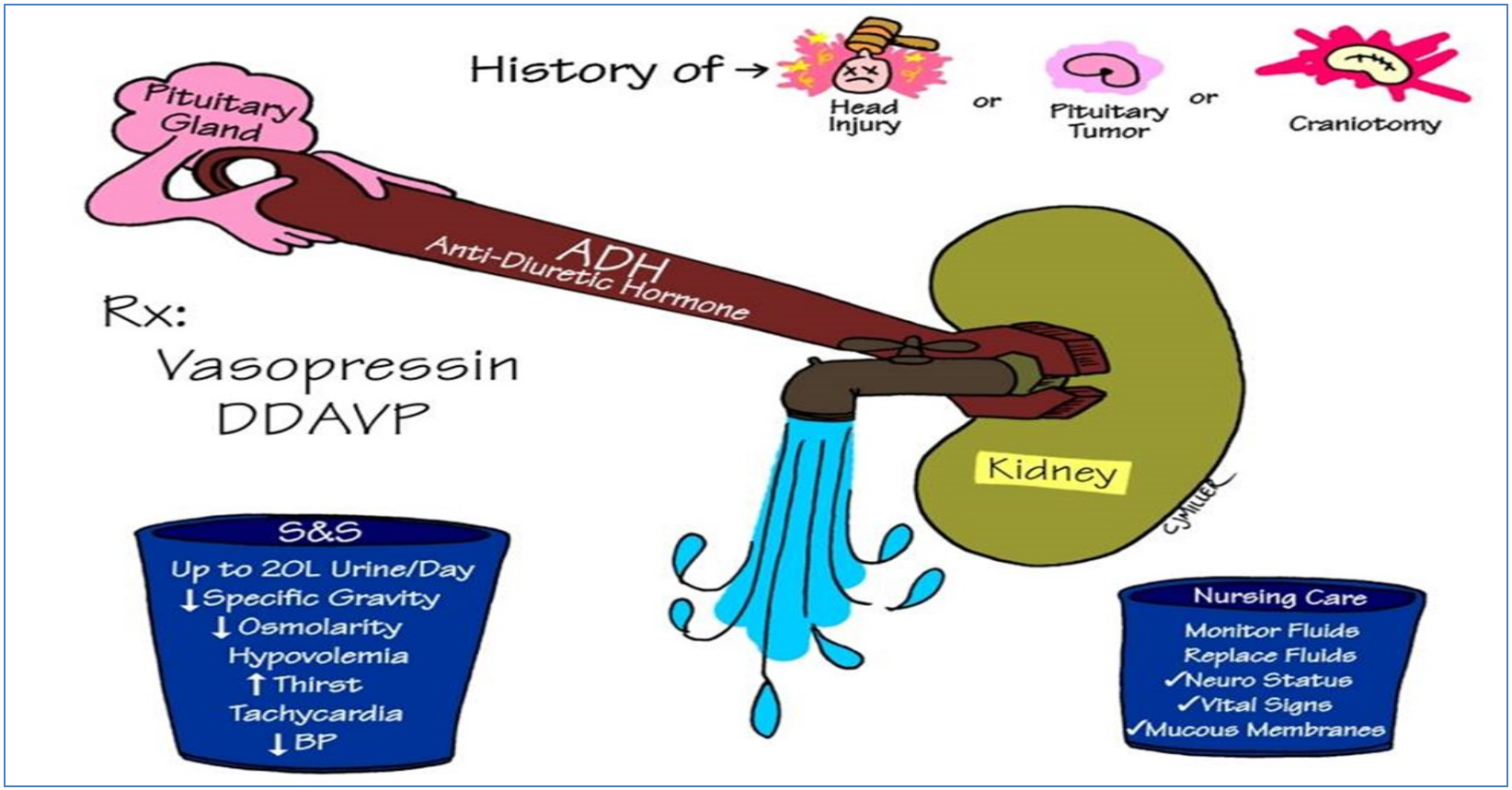
ADH secretion is stimulated by :	**ADH secretion is inhibited by :**
▪ Acetyl choline. ▪ Beta-adrenergic agonists. ▪ Chlorpropamide, Carbamazepine,Cyclophosphamide ▪ Nicotine, morphine.	▪ Diphenylhydantoin. ▪ Narcotic antagonists (e.g. Naloxone

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Diabetes Insipidus ‘DI’ - Tasteless Diabetes

Definition

- It is a disorder of impaired renal conservation of water which results from low or absent ADH response to normal physiological stimuli (central DI) or from renal unresponsiveness to ADH (nephrogenic DI).



Causes

➢ Central DI:

1- Familial : rare	2- Aquired
• Dominant inheritance. 'AD' • Recessive inheritance. DIDMOAD or 'Wolfram syndrome: o DI. o DM. o Optic atrophy o Nerve deafness	• Same causes of hypopituitarism

➢ Nephrogenic DI:

1- Familial	2- Acquired " 4D "
	• Disturbances of electrolytes. o Hypokalaemia e.g. lry aldosteronism. o Hypercalcemia e.g. hyperparathyroidism. • Drugs: e.g. lithium, De methylchlortetracycline. • Diseases of the kidney: e.g. o Chronic renal failure o Acute tubular necrosis & after transplantation. • Disorders (systemic disorders) :e.g. MM, amyloidosis, SCA

Clinical Picture

1.Polyuria: (urine volume 2.5 - 6 / 24 hr, may reach up to 20 L / day or more) → polydipsia & nocturnal enuresis in children.

2.Dehydration: may occur→weakness, prostration, low grade fever, psychic disturbances & death

Investigations

1.Urine examination: urine volume is ↑ with low specific gravity (1001- 1004) & fail to ↑ above 1010 after fluid deprivation.

2.X-ray sella turcica: to exclude tumours. [or CT/MRI].

4.Hypertonic Nacl test: (To confirm DI): IV. Nacl (in a dose of 2.5 cc 2.5 % solution / Kg). The result is stimulation of the osmoreceptors (if sensitive) → oliguria

3.Vasopressin test: (to differentiate causes): IV. infusion of vasopressin or DDAVP (10-25 mg S.C)→ for 2-3 days.

a. Normal subjects & patients with central D.I →oliguria.

b. Patients with nephrogenic DI. → no response.

5.Dehydration (water deprivation) test: (to differentiate causes): Steps:

- Deprive the patient of water for 4 - 18 hrs (with hourly measurement of urine osmolality).
- End point of the test :

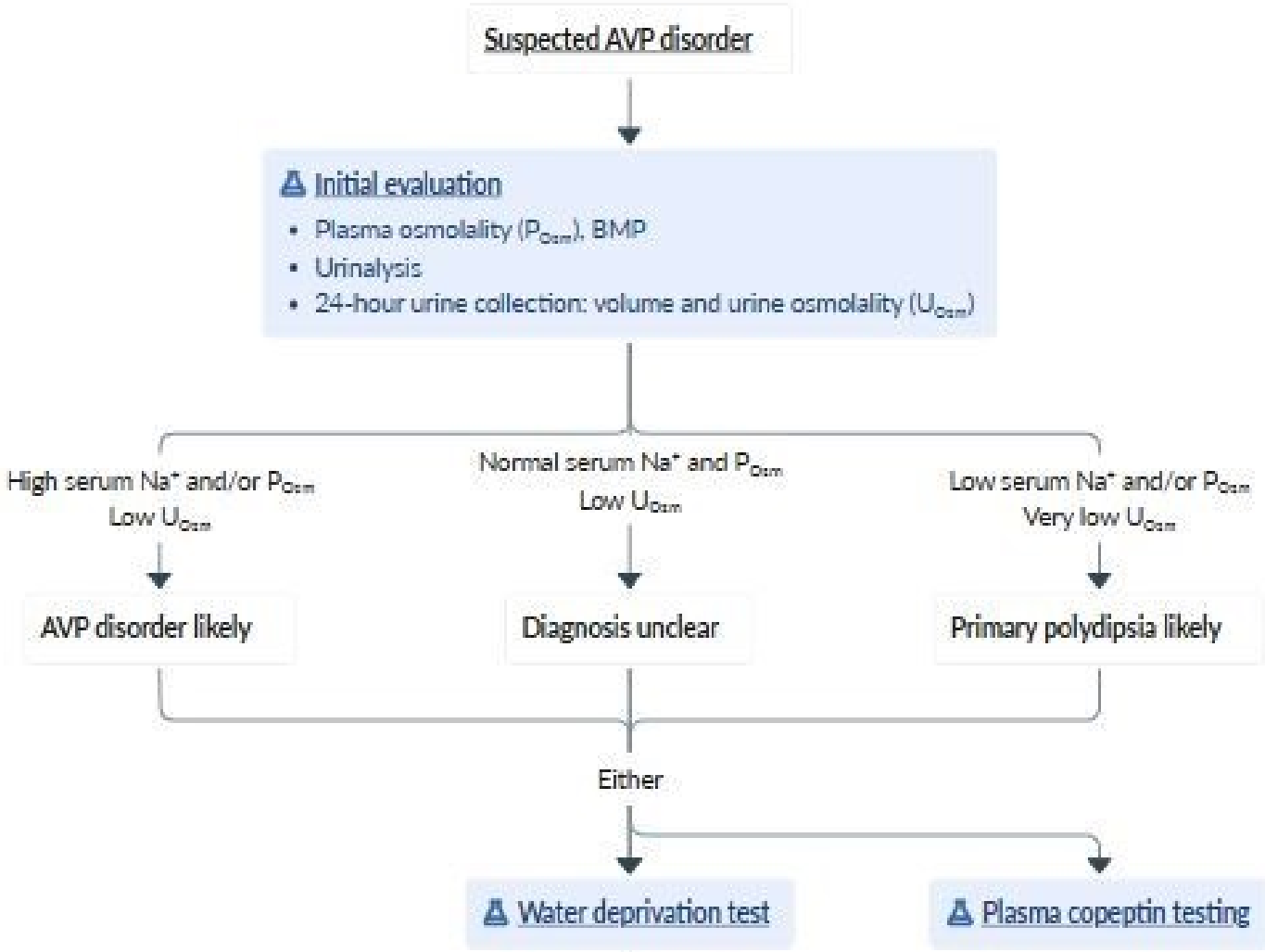
- Successive samples of urine of the same osmolality.

Or - Weight loss (5 % the body weight).

- Give 5 units of aqueous ADH S.C.& measure urine osmolality 1 hour later.

	Osmolality of urine	
	At the end point	After giving ADH
normal subjects & in psychogenic polydipsia	↑	[No further ↑] -
Central D.I.	- (low)	↑
Nephrogenic D.I.	- (low)	- (low)

Diagnostic approach to suspected arginine vasopressin disorder (diabetes insipidus)



Treatment

Drugs for central DI	Drugs for nephrogenic DI
Hormonal replacement therapy: • Aquous vasopressin (pitressin): SC • Desmopressin (DDVAP, des-amino, d-arginine vasopressin). Oral (minirin) or sublingualIt has the advantage of not causing pallor, angina, colic, bronchospasm or uterine contraction) • Vasopressin tannate in oil (pitressin) IM	• Hydrochlorothiazide or chlorthalidone: Diuretics→hyponatremia→↓amount of Na, delivered to the loops of henle→↓ water execretion. • Amiloride (potassium-sparing diuretic): is the drug of choice in lithium toxicity-induced DI
Non hormonal agents(↑ ADH) • chlorpropamide • clofibrate • carbamazepine	

Other lines of ttt:

- Treatment of the cause.
- Diet: o Liberal amounts of fluids are allowed.
  o Diet rich in purines should be avoided as it tends to exaggerate polyuria.

Differential diagnosis:other causes of polyuria (≥ 3 L / day)

➢ Renal causes :

Chronic renal failure (e.g. chronic pyelonephritis, chronic glomerulonephritis).

- Diuretic stage of acute tubular necrosis.
- Diuretics in excess.
- Nephrogenic DI.
- Renal tubular acidosis & Detoni - fanconi syndrome.

➢ Endocrine causes:

- Diabetes mellitus (specific gravity of urine is high, urine contians sugar & blood sugar level is high).
- Diabetes insipidus.
- Adrenal causes → Cushing's S. , conn's S. & addison's disease.
- Hyperparathyroidism(calcium diabetes)& other causes of hypercalcaemia.
- Thyrotoxicosis.

➢ Functional causes :

- Excessive intake of fluids, coffee, tea, cola, beer.. Etc.
- Hysterical (1ry psychogenic) polydipsia :
  - o Polyuria & polydipsia, no nocturia.
  - o Evident emotional disturbances e.g. anorexia nervosa.
  - o Low urinary osmolality with normal response to dehydration.
- During Winter.

➢ Miscellaneous causes:

- Multiple myeloma, Vit. D intoxication.
- Transient polyuria may occur after attacks of migraine, epilepsy, paroxysmal tachycardia, bronchial asthma, intermittent hydronephrosis.

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Syndrome of Inappropriate ADH Secretion ‘SIADH’

There is ↑ ADH secretion (which is not response to normal physiologic stimuli)→↓ ability to excrete ingested water)

Causes:

- Malignancy e. g. Oat cell carcinoma ‘of lung”.
- Non-malignant pulmonary diseases e.g. pneumonia, T. B.
- Cerebrovascular accidents.
- Drugs e.g. chlorpropamide cyclophosphamide Vincristine.
- Miscellaneous e g. hypothyroidism, acute phychosis.

Clinically: (water intoxication)

- **Symptoms of hyponatremia**: Chronic hyponatremia may be asymptomatic and CNS symptoms are less common.
- **Mild:** Anorexia, Nausea, vomiting, Headache, Muscle cramps.
- **Moderate:** Muscle weakness, Lethargy, Confusion.
- **Severe:** Seizures, Altered consciousness.
- **Other clinical features:** Normotension, No oedema
- **Symptoms of the underlying condition.**

SIADH patients are usually euvolemic, normotensive, and have no edema. A hyponatremic patient with edema should raise suspicion for other conditions (e.g. congestive heart failure).

Laboratory finding:

- Serum Na < 130 mEq/l (dilutional hyponatremia) (euvolemic hyponatremia).
- ↓ plasma osmolality (<270 mosmol/ kg)
- ↑ urine osmolality (> 100 mosmol/ kg)

Diagnostic criteria for SIADH [5][6][7][8]	
	Clinical and/or laboratory findings
Hyponatremia	• Serum sodium < 135 mEq/L
Hypoosmolality	• Serum osmolality < 275 mOsm/kg
Euvolemia	• No signs of hypovolemia • No signs of hypervolemia
Concentrated urine	• Urine osmolality > 100 mOsm/kg
Elevated urinary sodium	• Urine sodium concentration > 20–30 mEq/L

Treatment:

- Of the cause
- Restrict water intake.

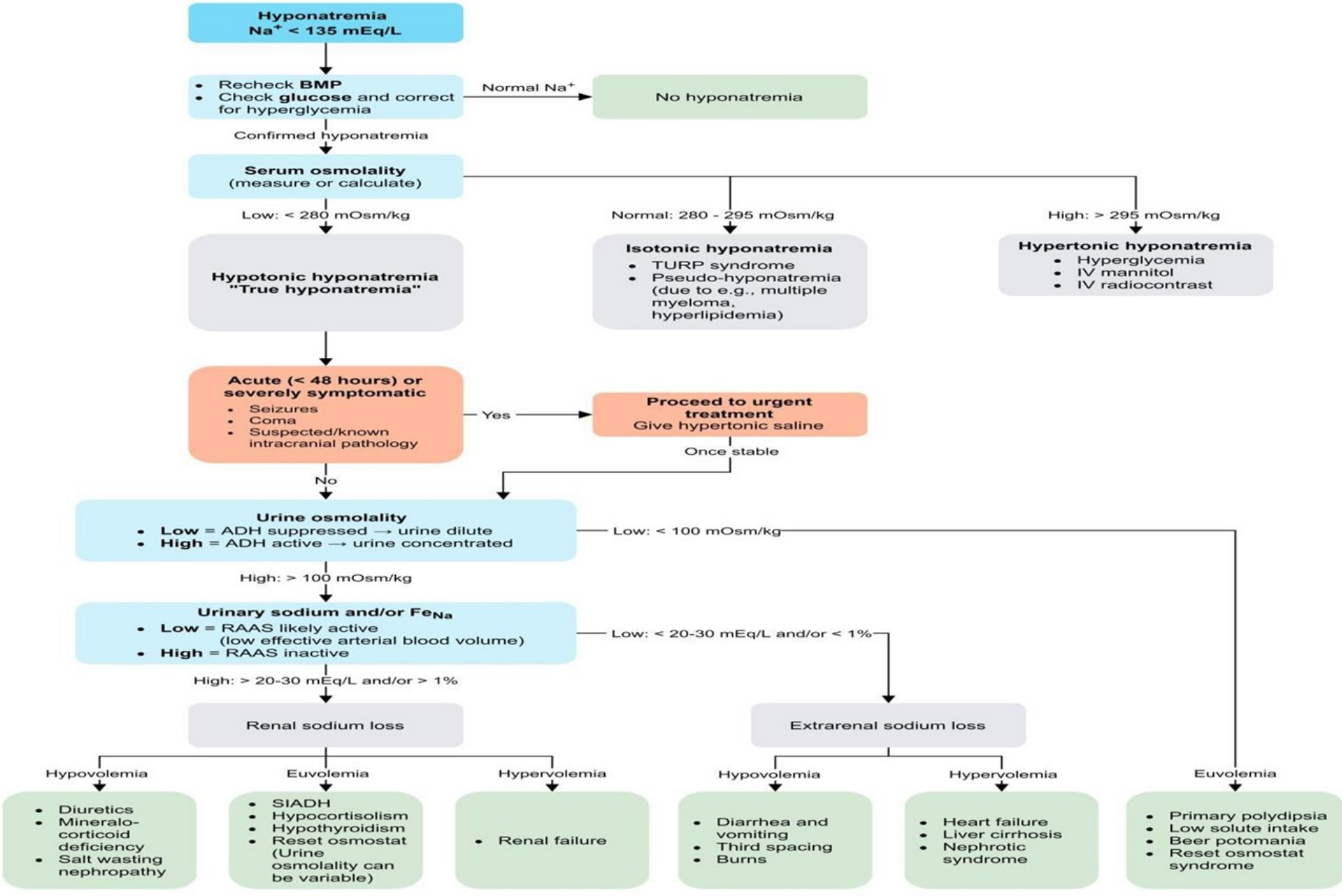
> **Clinical pearl**
>
> Ideally, daily fluid intake should be 500 mL less than daily urine output.

- Drugs:
  - o demethylchlotetracycline (demeclocycline).
  - o Diphenylhydantoin.
  - o Naloxone.
- Hypertonic saline infusion (5 %) in emergency.

> **Clinical pearl**
>
> For patients with chronic hyponatremia, the maximum limit (not goal) for serum sodium level increase is 10–12 mEq/L per 24 hours. Use a lower maximum limit of 8 mEq/L per 24 hours for patients with high-risk factors for osmotic demyelination syndrome. 7

> **Caution**
>
> In patients with chronic hyponatremia, sodium overcorrection can lead to **osmotic demyelination syndrome**. 5 7

Diagnostic approach to hyponatremia



- A 63-year-old woman comes to the physician because of a 4-week history of increased urinary frequency, bone aches, nausea, and fatigue. She says she has felt increasingly thirsty despite drinking 5–6 liters of water daily and has been waking up frequently during the night to urinate. The patient has type 2 diabetes mellitus, major depressive disorder, and hypertension. Current medications include lisinopril, escitalopram, and metformin. She drinks one beer each night. She is 165 cm (5 ft 5 in) tall and weighs 80 kg (176 lb); BMI is 29 kg/m2. Her vital signs are within normal limits. Examination shows dry mucous membranes. Laboratory studies show:

Hemoglobin	14.3 g/dL
Serum	
Na^+	151 mEq/L
K^+	3.9 mEq/L
Cl^-	102 mEq/L
Ca^{2+}	13 mg/dL
Urea nitrogen	8 mg/dL
Creatinine	1.0 mg/dL
Glucose	101 mg/dL
Osmolality	306 mOsmol/kg H_2O
Parathyroid hormone	130 pg/mL
Urine	
Osmolality	255 mOsmol/kg H_2O

Following a water deprivation test, plasma osmolality is 309 mOsmol/kg H2O and urine osmolality is 256 mOsmol/kg H2O. One hour after desmopressin administration, plasma osmolality is 305 mOsmol/kg H2O and urine osmolality is 259 mOsmol/kg H2O.

Which of the following is the most likely cause of this patient's increased urinary frequency?

a) Nephrogenic diabetes insipidus
b) Hyperosmolar hyperglycemic state
c) Beer potomania
d) Psychogenic polydipsia
e) Central diabetes insipidus
f) Syndrome of inappropriate ADH secretion

Summary



References

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Thank You